C1.5 Transaction processing

When a master component issues a transaction, the coherency signaling indicates whether the transaction is to a memory location that is shareable by more than one component, and therefore requires coherency support. Typically, transactions are processed as follows:

1. The initiating master component issues the transaction.
2. Depending on whether coherency support is required, either:
   - the transaction is passed directly to a slave component, subject to the address decode scheme being used
   - the transaction is passed to the coherency support logic within the interconnect.
3. A coherent transaction is checked against subsequent transactions from other master components, to ensure correct processing order.
4. The interconnect determines the snoop transactions that are required.
5. Each cached master that receives a snoop transaction must provide a snoop response. Some masters might provide snoop data to complete the snoop transaction.
6. The interconnect determines whether a main memory access is required.
7. The interconnect collates snoop responses and any required data.
8. The initiating master component completes the transaction.

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